Reasoning-Based Weekly Meal Planner for Safe, Constraint-Aware Meals

Deterministic Multi-Agent System for Allergies, Halal/Kosher, and Vegan Constraints

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Problem Statement

People with severe food allergies or strict dietary rules (halal, kosher, vegan) must carefully inspect every recipe before cooking. A single hidden allergen, ambiguous ingredient, or unsafe substitution can make a dish dangerous.

Despite the popularity of recipe websites and AI assistants, these tools cannot reliably guarantee safety. Recipe apps rely on shallow keyword filters, and AI systems may hallucinate ingredients or overlook violations.

As a result, meal planning remains manual, time-consuming, and error-prone for those who need strict dietary compliance.

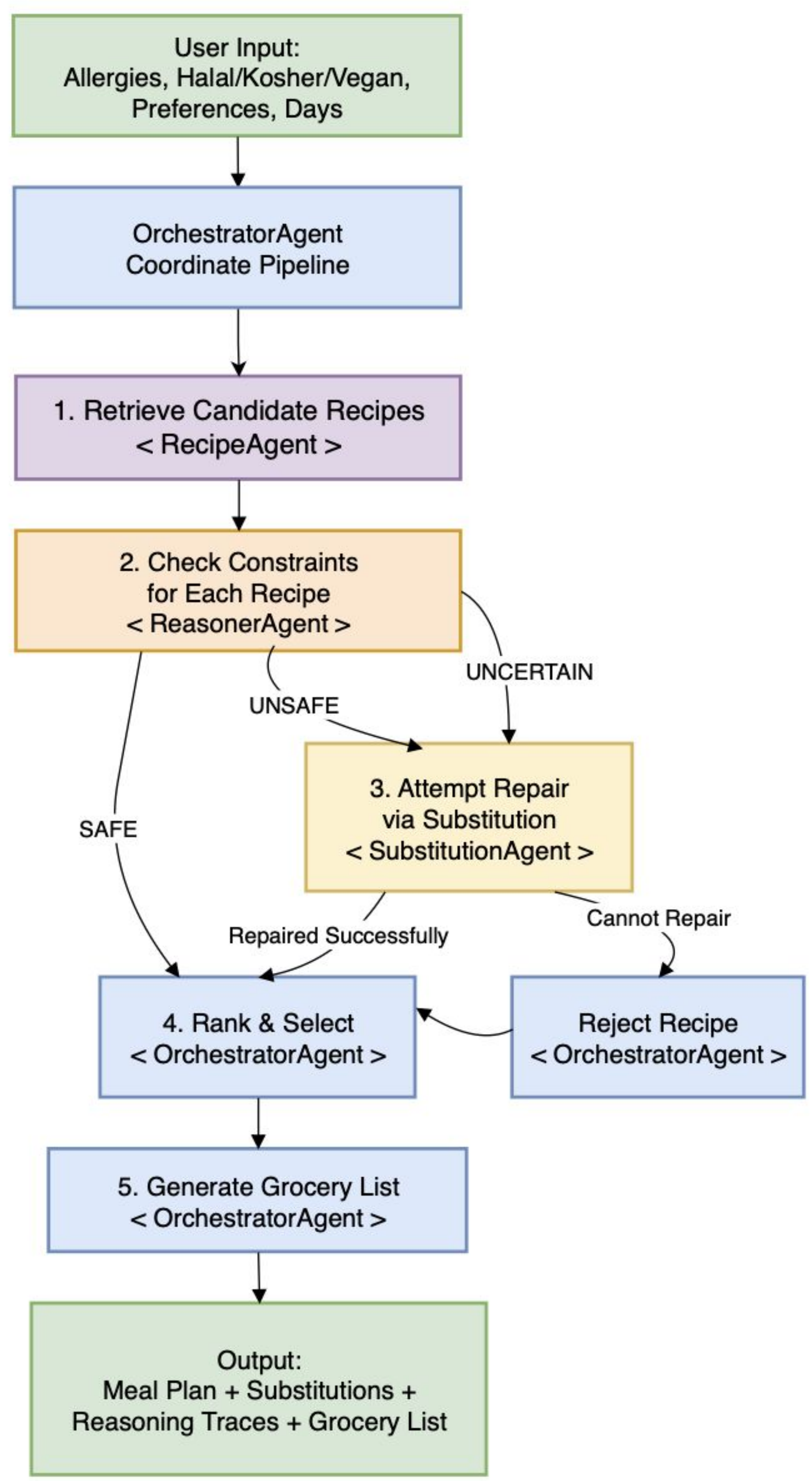
Solution

We present a system that creates weekly meals, satisfying dietary constraints such as allergies, halal/kosher, and veganism, as hard constraints. Unlike other systems that rely on keyword searches and probabilistic AI outputs, our system uses structured checks on each ingredient to satisfy these constraints.

In cases where a recipe is considered unsafe, our system automatically corrects it by replacing ingredients with safe ones, allowing it to be as similar to the original recipe as possible. In cases where it is impossible to safely correct a recipe, it is rejected.

The system provides a deterministic reasoning trace for every decision made, allowing users to verify whether all dietary constraints have been satisfied.

Methodology



Representation

- **User profile:** allergies + halal/kosher/vegan + manual avoid (hard), plus soft preferences and requested days.
- **Recipe + knowledge:** recipe corpus (title/ingredients/instructions) + allergen DB + deterministic dietary rules + risk lexicon + substitution graph.

Multi-Agent Reasoning Pipeline

1. **Retrieve candidates (RecipeAgent)**
   Recall-oriented retrieval from the recipe corpus using optional keywords/preferences.
2. **Deterministic constraint checking (ReasonerAgent)**
   Ingredient-level checks produce **SAFE / UNSAFE / UNCERTAIN** with an auditable trace:
   title guard → manual avoid → ambiguity → risk lexicon → allergen DB → halal/kosher/vegan rules.
   UNCERTAIN is treated explicitly (no "guessing safe").
3. **Repair loop (SubstitutionAgent + re-check)**
   For UNSAFE/UNCERTAIN recipes, the system replaces violating ingredients using the substitution graph, then re-runs the deterministic checks.
   If repair succeeds → keep; otherwise → reject.
4. **Rank & select (OrchestratorAgent)**
   Among safe/repaired recipes, apply soft constraints **only for ranking** (never overriding safety), then select recipes to fill the week.
5. **Output**
   Weekly meal plan + grocery list + **reasoning traces** and **substitution logs** for transparency.

RESULTS

System Output

- Generates N-day meal plan and shopping list
- For each recipe:
  - label (VERIFIED / REPAIRED / UNSAFE / UNCERTAIN)
  - explanation + trace (auditability)

Key Behaviors

- No unsafe output: only recipes passing the final safety gate are returned.
- Uncertainty-aware: ambiguous ingredients trigger UNCERTAIN instead of unsafe guesses.
- Automatic repair: when a recipe fails, system attempts substitution and re-checks.

Demo Example

CONCLUSION

Our system shows that safe meal planning under strict dietary constraints is best treated as a constraint-satisfaction and repair problem rather than pure text generation. A deterministic, ingredient-level reasoner classifies recipes as SAFE/UNSAFE/UNCERTAIN, applies substitutions when possible, and rejects recipes that cannot be made compliant. Each output includes a clear reasoning trace and (when repaired) a substitution log, making decisions auditable and consistent. This design produces more reliable results than LLM-only planners while allowing bounded LLM use for late-stage normalization and verification.

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